Project 5 — Game Design Document

Nov 9, 2026

Overview

A top-down roguelite bullet-hell built in Godot 4: fight through 8 locked rooms, graze enemy bullets to charge a Shockwave, and beat the boss in room 8. Team: K'von Gates, Daryl Watkins-Mattocks, Jarman. Working title: not chosen yet.

Assumptions this draft makes (change any of them and the numbers below follow):

- **Engine and units.** Godot 4, 2D scenes with characters pre-rendered from 3D models (see Art direction). Every distance is in pixels. The kickoff's 3 m barrel radius converts at 1 m = 32 px, so 96 px.
- **Proposed numbers.** Where the kickoff gave none (player HP, weapon, enemy HP, room count), this draft proposes values and marks them *(proposed)*.
- **Jarman's mechanics** were marked Placeholder at kickoff. They are treated as final here.
- **K'von's second mechanic** is Room-Based Combat. Randomized Dungeon Generation is folded into it as a rule, not a separate mechanic (see Scope and cuts).

Core loop

The player **dodges and grazes enemy bullets**, in order to **fill the Graze Meter and fire Shockwave**, which lets them **clear the room's waves and unlock the next room**, and then **dodges and grazes enemy bullets** again.

The kickoff's version ("dodges and grazes … repeat until they reach and defeat the final boss") is tightened to the X / Y / Z / X form the brief asks for; reaching the boss is covered under Win and lose.

Mechanics

Six mechanics, two per member. The player baseline sets the scale the other numbers work against.

Mechanic	Owner	Input

Environmental Combat	K'von Gates	WASD move, mouse aim, left-click to shoot objects
Room-Based Combat	K'von Gates	WASD to walk through a door into the next room
Bullet Grazing	Daryl Watkins-Mattocks	WASD to move close to enemy bullets
Shockwave	Daryl Watkins-Mattocks	Q, when the Graze Meter is at 100
Enemy Adaptation	Jarman	Automatic, enemies react to the room
Combat Rewards	Jarman	Automatic, on each enemy defeat

Player baseline *(proposed)*

Rule	Number
Player HP	100
Move speed	220 px/s
Player shot damage / fire rate	10 per shot / 6 shots per s (hold left-click)
Enemy bullet damage	10 (10 hits from full HP)
Enemy HP: standard / elite / boss	40 / 120 / 1,200

1. Environmental Combat — K'von Gates

Input: WASD to move, mouse to aim, left-click to shoot an object; traps trigger when stepped on.

Rule	Number
Interactive objects per room	1–3, type and spot picked at random each run
Explosive barrel	Detonates after 2 hits; 30 damage to enemies within 96 px (3 m)
Destructible cover	Blocks enemy bullets; breaks after 5 hits from any source
Trap	Triggers when stepped on: 15 damage to the enemy or player on it; re-arms after 2 s *(proposed)*

2. Room-Based Combat — K'von Gates

Input: WASD to walk through a door. Crossing the threshold starts the room.

Rule	Number
Rooms per run	8: rooms 1–6 combat, room 7 elite, room 8 boss *(proposed)*
Layouts	Drawn at random from a pool of 10 hand-built layouts, no repeats in a run; objects placed on 3–6 spawn markers per layout
Doors	Lock on entry; unlock when the last wave is defeated
Waves	Rooms 1–3: 2 waves; rooms 4–6: 3 waves; each wave 4–6 standard enemies; from room 4, 1 elite per wave
Elite room (7)	1 wave: 2 elites + 4 standard
Wave timing	Next wave spawns 1.5 s after the previous one is cleared
Spawn distance	Enemies spawn at least 200 px from the player
Room-clear heal	+10 HP, capped at 100 *(proposed)*

3. Bullet Grazing — Daryl Watkins-Mattocks

Input: WASD to move close to enemy bullets without being hit.

Rule	Number
Graze distance	Bullet passes within 20 px of the player hitbox without hitting
Meter gain per graze	+5
Graze Meter maximum	100 (20 grazes without damage fills it)
Repeat grazes	Each bullet can be grazed once
Taking damage	−25 meter

4. Shockwave — Daryl Watkins-Mattocks

Input: Q, only when the Graze Meter is at 100.

Rule	Number
Cost	All 100 meter points
Radius	150 px around the player
Effect on bullets	Destroys every enemy bullet in the radius
Effect on enemies	30 damage and 50 px knockback
Activation	1 s animation; player is invulnerable for that second

5. Enemy Adaptation — Jarman

Input: none; enemies respond to the room's layout and hazards.

Rule	Number
Obstacle detection	Within 200 px
Pathing	Routes around walls and destructible cover while chasing the player
Hazard avoidance	Steers clear of active hazards within 75 px
Path recalculation	Every 2 s
Ranged enemy distance	Keeps 150–250 px from the player, behind cover when cover is in range

6. Combat Rewards — Jarman

Input: none; triggers on each enemy defeat.

Rule	Number
Energy per kill	Standard 5, elite 15, boss 30; added to the Graze Meter up to 100
Damage power-up drop	Every 5th enemy defeated
Power-up effect	+20% weapon damage for 10 s
Stacking	Does not stack; a new pickup resets the timer to 10 s

How they plug in

Each member's two mechanics change each other, and one of each member's mechanics changes a teammate's. Rules marked *(new)* were added so the link has a number behind it.

Member	Mechanic	Changes	How
K'von	Environmental Combat	Room-Based Combat (own)	The 1–3 objects in a room decide how its waves play: a barrel can wipe a clustered spawn, cover splits the room into safe lanes.
K'von	Room-Based Combat	Environmental Combat (own)	Each layout's 3–6 spawn markers decide where objects can appear, and every new room rolls new objects, so no two runs use the same setups.
K'von	Environmental Combat	**Combat Rewards (Jarman)**	*(new)* An enemy killed by a barrel or trap pays double energy (10 / 30 / 60) and still counts toward the every-5th-kill power-up.
Daryl	Bullet Grazing	Shockwave (own)	Grazing is the main way to reach the 100 points Shockwave needs: 20 clean grazes = one Shockwave.
Daryl	Shockwave	Bullet Grazing (own)	Firing empties the meter to 0 and deletes every bullet within 150 px, so the player has to move back into fire to graze again.
Daryl	Shockwave	**Environmental Combat (K'von)**	*(new)* Barrels inside the 150 px radius detonate at once, and the 50 px knockback can push enemies onto traps. Shockwave clears bullets the way cover blocks them, so the player can trade cover for a Shockwave.
Jarman	Enemy Adaptation	Combat Rewards (own)	Enemies steer clear of hazards within 75 px, so the double-energy environmental kills only happen when the player lures or knocks enemies in.
Jarman	Combat Rewards	Enemy Adaptation (own)	*(new)* While the +20% damage boost is active, ranged enemies back off to 200–300 px and recalculate paths every 1 s instead of 2 s.

Jarman	Enemy Adaptation	**Room-Based Combat (K'von)**	Enemies read each room's layout: ranged enemies take positions behind cover 150–250 px from the player, so the same wave plays differently in each of the 10 layouts.

One more cross-link carries over from kickoff: Combat Rewards adds kill energy to Daryl's Graze Meter, a second way to charge Shockwave.

Win and lose

- **Win:** defeat the boss in room 8 (1,200 HP *(proposed)*). The run ends with a victory screen.
- **Lose:** player HP reaches 0 in any room. The run ends and the player restarts at room 1 with a new random set of layouts and objects. HP, Graze Meter and damage boost reset; nothing carries over between runs.

Art direction

Characters, enemies and props are low-poly 3D models with hand-painted textures, rendered from one fixed 3/4 top-down camera into 2D sprite sheets, the way Hades gets a painted-2D look from 3D. Godot then runs as a 2D game, so all the pixel numbers above hold.

Pipeline: model and texture in Blender → render with an orthographic camera at a fixed 3/4 angle, key light from the top-left → 8-direction sprite sheets → Godot `AnimatedSprite2D`. Floors and walls are painted 2D tiles from the same angle.

Palette

One rule above the rest: Hostile Magenta appears on enemy bullets and nothing else, so bullets always read against the room.

Swatch	Hex	Used for
Abyss	`#120E1A`	Void, background, ink outlines
Crypt Stone	`#2B2335`	Floor tiles
Stone Light	`#4E4458`	Walls, destructible cover

Bone	`#D8C8A8`	UI text, highlights
Ember	`#F06A2A`	Player accents, explosive barrels, fire
Blood Rust	`#8E2A2A`	Traps, damage flashes
Graze Gold	`#FFC93C`	Graze Meter, graze sparks, Shockwave ring
Spirit Teal	`#35D6C4`	Player shots, damage power-up
Hostile Magenta	`#FF2E6E`	Enemy bullets only

Style suffix

Appended to the end of every image or concept prompt so all assets come out matching:

```
None
, 3D model with hand-painted textures rendered as a 2D game sprite, fixed 3/4 top-down orthographic view, bold dark ink outlines, cel-shaded with painterly brushstroke highlights, strong top-left key light and warm rim light, dark gothic dungeon mood, limited palette of deep violet, ember orange and gold, clean readable silhouette, plain background
```

Avoid list

- Photorealism, and pixel art: both break the painted-3D look
- Perspective cameras or any angle other than the fixed 3/4 view
- Chibi or cartoon proportions
- Magenta or pink on anything except enemy bullets
- Bloom, glow or particle effects that hide bullets or the player hitbox
- Busy, high-contrast floor textures where bullets travel
- Text, logos or watermarks inside generated images
- Copying Hades characters, enemies or UI: match the technique, not the designs

Anchor asset

The **player character's idle sprite**, facing down-right, about 64 px tall in a 128 × 128 frame. Every other asset is checked against it for scale, outline weight, light direction and palette before it goes in the game. It is drawn first, by November 18.

Architecture

Drawn before building: one box per script, labelled with the mechanic state it holds, and an arrow from each script to every script it reads. The code will drift from this, and that is fine.

Each mechanic’s state lives in one scriptGodot 4 scripts · arrow points from a script to the script it reads · two heads = both read each otherHUD.gdHUD.tscnNo mechanic stateDraws HP, meter, room, boostGameManager.gdMain.tscn, autoloadWin and lose: run state,room index 1–8Player.gdPlayer.tscnHP 0–100, move, aim, shoot,damage multiplierRoomManager.gdRoom.tscnRoom-Based Combat:layout, wave, enemies alive,doors lockedGrazeMeter.gdchild of Player.tscnBullet Grazing:meter 0–100,grazed bullet IDsCombatRewards.gdautoloadCombat Rewards:kill count (every 5th),boost timer 0–10 sEnemy.gdEnemy.tscnHP, type, killed-by sourceEnemy Adaptation: path,2 s timer, hazards seenEnemyBullet.gdEnemyBullet.tscndamage 10, grazed flagShockwave.gdchild of Player.tscnShockwave: ready flag,1 s invulnerable timerEnvObject.gdBarrel, Cover, Trap scenesEnvironmental Combat:type, hits taken (0–5),trap armedColour = mechanic ownerK’von GatesDaryl Watkins-MattocksJarmanShared support

Player.gd and Enemy.gd are the hubs: five scripts read the player, four read enemies. The one long arrow on the right is EnvObject reading the player for trap and blast damage.

Script	Reads	For
GameManager.gd	RoomManager, Player	Room cleared → next room; HP 0 → lose
HUD.gd	GameManager, Player, GrazeMeter	Room number, HP and boost, meter fill
Player.gd	CombatRewards	Damage multiplier while boosted
RoomManager.gd	Enemy	Enemies alive → next wave or unlock doors
GrazeMeter.gd	EnemyBullet, Player	Grazed flag and distance; hits taken (−25)
Shockwave.gd	GrazeMeter, EnemyBullet, Enemy, EnvObject	Meter = 100; bullets, enemies and barrels in 150 px

Enemy.gd	Player, EnvObject, CombatRewards	Chase target; cover and hazards in 200 px; boost active
EnvObject.gd	Enemy, Player	Who is in the blast or on the trap
CombatRewards.gd	Enemy, GrazeMeter	Enemy type and killed-by source; add energy up to 100

Scope and cuts

Each owner builds their own mechanics. Nov 18 gets the core loop playable in one room: move, shoot, graze, Shockwave, waves, kill energy. Dec 2 adds the systems that depend on room layouts.

Mechanic	Builder	Works by
Bullet Grazing	Daryl Watkins-Mattocks	Nov 18
Shockwave	Daryl Watkins-Mattocks	Nov 18
Room-Based Combat	K'von Gates	Nov 18 (one layout, waves, door locks); all 10 layouts by Dec 2
Combat Rewards	Jarman	Nov 18
Environmental Combat	K'von Gates	Dec 2
Enemy Adaptation	Jarman	Dec 2

Support scripts *(proposed builders)*: Player.gd and HUD.gd by Daryl, Enemy.gd base and EnemyBullet.gd by Jarman, GameManager.gd by K'von, all by Nov 18. The boss fight is due Dec 2. Open question: who draws the anchor asset.

Cuts

- **Dropped at kickoff:** None.
- **Dropped since:** Randomized Dungeon Generation as its own mechanic. It is now one rule of Room-Based Combat: rooms are drawn from 10 hand-built layouts instead of generated procedurally, which fits a 3-week build.